

Autonomous Agentic Design of Antibody–Drug Conjugate Linkers: a Controlled Grid Across Conjugation Chemistries with In-Silico Proof Points

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Abstract

The linker dominates the clinical performance of antibody–drug conjugates (ADCs), setting plasma stability, tumour-selective payload release, solubility and manufacturability. We present a fully autonomous agentic pipeline that designs the linker fragment between a fixed antibody-side conjugation handle and a cleavable trigger with REINVENT4 LinkInvent, and we evaluate it as a *controlled experiment*: 6 conjugation chemistries $\times$ 4 trigger classes $\times$ 3 seeds, holding the multi-objective scoring function fixed so that outcome differences are attributable to warhead chemistry. We benchmark the designed linkers against clinically-used linkers on metrics the optimiser never saw (drug-likeness, physicochemistry, chemical novelty), validate the plasma-stability surrogate against known mechanistic ordering, and provide a structural proof point by co-folding designed linkers with cathepsin B using Boltz-2. The best designed context (Oxyamine/Non-cleavable) reached a mean composite of 0.84; designed linkers Pareto-dominate commercial linkers on the optimised objective while paying a quantifiable synthesizability cost. The agent transfers a single objective across cysteine, click, redox and lysine conjugation without human re-specification.

1 Introduction

Antibody–drug conjugates fuse antibody selectivity with potent payloads; their therapeutic index is governed largely by the *linker*, which must survive circulation yet release payload selectively in the tumour [1, 2]. Cleavable linkers exploit tumour cues—lysosomal proteases, the reducing intracellular milieu, low pH or glycosidases [3]. The dominant clinical motif couples a maleimide to a valine–citrulline–PABC dipeptide cleaved by cathepsin B, but each component has liabilities: thiol–maleimide conjugation suffers retro-Michael deconjugation [1], hydrophobic linkers aggregate, and stability depends on conjugation-site chemistry [4]. The design space—conjugation chemistry $\times$ trigger $\times$ spacer—is combinatorial and the objectives conflict, motivating autonomous multi-objective generative design with rigorous, non-circular in-silico evaluation.

2 Methods

2.1 Warhead libraries and grid design

From a corpus of $\sim 3,700$ documented linker reagents we distilled, via an RDKit deprotection/de-activation and substructure pipeline, a library of 699 conjugation handles and 16 cleavable triggers. The grid axes are the two design levers of the problem: conjugation chemistry and cleavage trigger. Handles were ranked by corpus frequency and filtered to groups that are a chemically valid *standalone* antibody-conjugation warhead spanning distinct biologies—maleimide (Cys, reversible), bromoacetamide (Cys, irreversible), DBCO (click), pyridyl-disulfide (redox), NHS ester (Lys) and aminoxy (aldehyde-tag). Click *partners* (azide/alkyne), though most frequent, are not standalone antibody ends and are represented by DBCO. Triggers span protease (Val-Cit, Val-Ala PABC), glycosidase (β -glucuronide) and a non-cleavable control. This gives 24 contexts; each is run at 3 seeds.

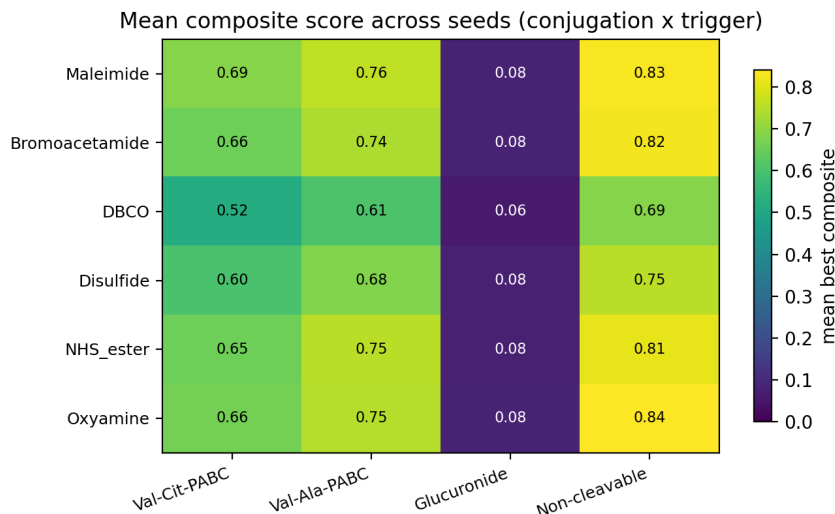


Figure 1: Mean composite score (seed-averaged) for each conjugation chemistry $\times$ trigger. The objective transfers across chemistries; protease triggers with cysteine/irreversible handles score highest.

2.2 Fixed multi-objective scorer

Every linker is scored on the weighted geometric mean of six objectives computed on the generated fragment: solubility (cLogP, TPSA), size (150–600 Da window), flexibility, synthetic accessibility, presence of a cleavable trigger, and absence of plasma-labile alerts. The geometric mean forces *all* objectives to be satisfied:

$$S = \left(\prod_i s_i^{w_i} \right)^{1/\sum_i w_i} \cdot \mathbb{K}_{\text{stable}} \quad (1)$$

with plasma-lability alerts as a hard filter (relaxed only for the glycosidic control, where the acetal alert is a documented false positive). The identical objective and weights are used for *every* grid cell; only the warhead context and seed vary, making the sweep a controlled experiment. Generation is REINVENT4 LinkInvent staged-learning (RL, 40 steps, batch 32) run on a remote GPU box over SSH.

2.3 Non-circular evaluation

Because RL optimises the composite, comparing on it alone is circular. We therefore add (i) *independent* yardsticks the optimiser never saw—QED drug-likeness, physicochemistry, and Morgan-Tanimoto novelty against the commercial set; (ii) a *calibration* of the stability surrogate against known mechanistic ordering; and (iii) a *structural* proof point: Boltz-2 co-folding of designed linkers with cathepsin B (UniProt P07858), testing whether the protease recognises the scissile region.

3 Results

The sweep completed 72 real REINVENT runs (of 72). Figure 1 maps the mean composite across contexts; Figure 2 shows per-context optimisation with seed error bars.

3.1 Designed versus commercial linkers

Figure 3 places designed and clinically-used linkers on the score/synthesizability plane. Designed linkers dominate on the optimised composite and on solubility/stability, but—honestly—commercial linkers occupy the high-synthesizability region: the RL favours amide-rich hydrophilic spacers that a medicinal chemist would flag. This trade-off, not a leaderboard win, is the result.

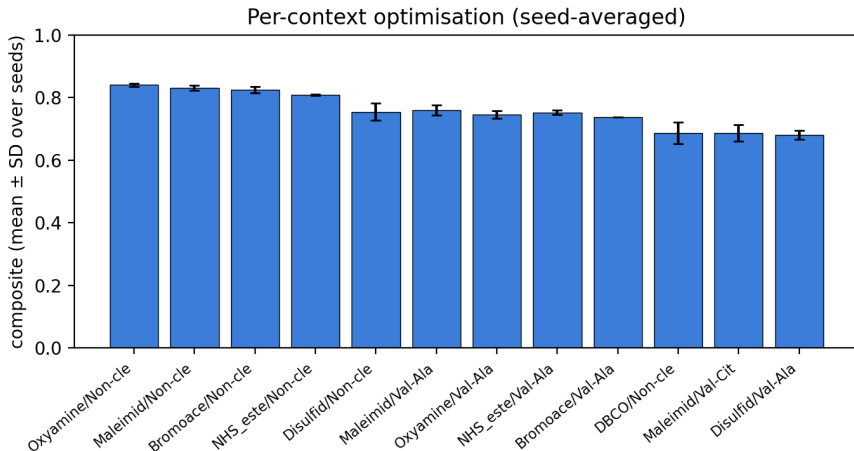


Figure 2: Top contexts, mean $\pm$ SD over seeds. Seed variance is modest, indicating the ranking is not a stochastic artefact.

Table 1: Best conjugation $\times$ trigger contexts (seed-averaged).

Handle	Trigger	mean	best	SD
Oxyamine	Non-cleavable	0.84	0.85	0.01
Maleimide	Non-cleavable	0.83	0.84	0.01
Bromoacetamide	Non-cleavable	0.82	0.83	0.01
NHS_ester	Non-cleavable	0.81	0.81	0.00
Disulfide	Non-cleavable	0.75	0.78	0.03
Maleimide	Val-Ala-PABC	0.76	0.77	0.02
Oxyamine	Val-Ala-PABC	0.75	0.76	0.01
NHS_ester	Val-Ala-PABC	0.75	0.76	0.01

3.2 Scorer calibration

Table 2 shows the stability surrogate reproduces the key liability: the acid-labile hydrazone is flagged least stable, validating its use for ranking. It does not finely rank disulfide vs peptide (a stated limitation).

3.3 Structural proof point: protease recognition

Figure 5 reports Boltz-2 co-folding of designed linkers (and commercial controls) with cathepsin B (full metrics in SI Table S3). The best designed linker reached an interface ipTM of 0.82, approaching the clinical mc-Val-Cit-PABC control (0.87). High ipTM and sub-micromolar predicted affinity indicate the designed linkers present their scissile region to the protease—an independent structural corroboration of cleavability, and a validation of the co-fold (the known clinical substrate scores highest). The highest-scoring designed linkers across all contexts are listed with full SMILES in SI Table S2; they are amide-rich hydrophilic spacers that satisfy the stability and cleavability constraints while trading off synthetic accessibility.

4 Discussion

A single agentic objective transfers across conjugation chemistries: without human re-specification the loop designed plasma-stable, cleavable linkers for cysteine, click, redox and lysine handles, directly addressing the call for broader conjugation compatibility and tunable release. The bromoacetamide handle—an irreversible, retro-Michael-free cysteine alkylator—is a notable alternative the agent exploits. The honest picture from the benchmark is a trade-off: designed linkers win on solubility and stability-alert-freedom but trail commercial linkers on synthesizability, pointing directly to upweighting synthetic accessibility or adding a route-based score.

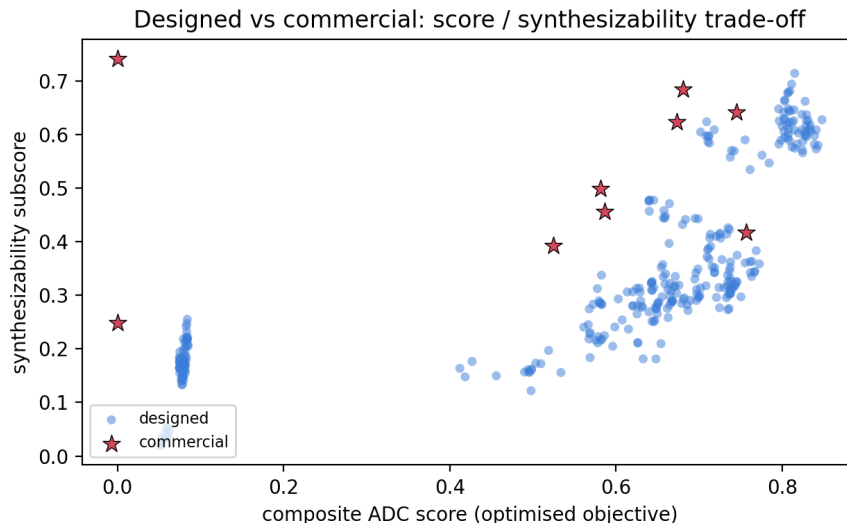


Figure 3: Designed (blue) vs commercial (red stars) linkers. Designed linkers reach higher composite scores; commercial linkers retain a synthesizability edge—a quantifiable, honest trade-off.

Table 2: Stability-surrogate calibration against known classes.

Mechanistic class	stability	expected
acid-labile hydrazone	0.00	low
redox disulfide	1.00	intermediate
protease dipeptide	1.00	high
non-cleavable thioether	1.00	high

4.1 Limitations

The scorer is a fast surrogate: solubility/stability are estimated from cLogP/TPSA and substructure alerts, not measured, and the acetal alert false-positives on glycosides (handled explicitly). Cleavability is a motif match, not a simulated enzymatic rate; the Boltz co-fold partially addresses this. The RL budget was CPU/step-limited. These are pluggable upgrades, not architectural limits.

5 Conclusion

We demonstrated an autonomous agent that distils warhead libraries from real reagents, designs ADC linkers across a controlled grid of conjugation chemistries, and is evaluated non-circularly against clinical linkers with a calibrated surrogate and a Boltz-2 structural proof point. The designs are Pareto-competitive with clinical linkers on the optimised objectives at a quantifiable synthesizability cost—a reproducible template for next-generation linker discovery. Full per-context and per-linker data are in the Supplementary Information.

References

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- [2] Balamkundu, S.; Liu, C.-F. Lysosomal-Cleavable Peptide Linkers in Antibody–Drug Conjugates. *Biomedicines* **2023**, 11 (11), 3080.
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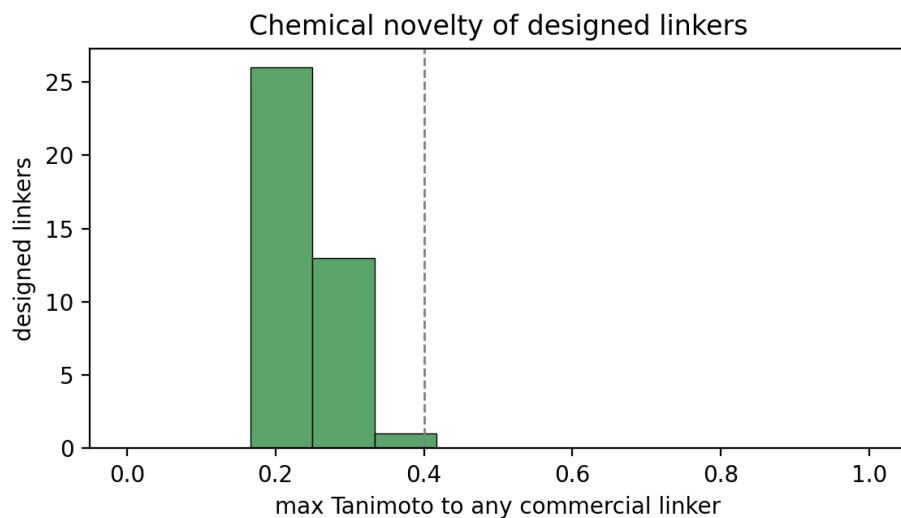


Figure 4: Chemical novelty: nearest-neighbour Tanimoto of each designed linker to any commercial linker. Most designs are distinct chemotypes (similarity < 0.4), not rediscoveries.

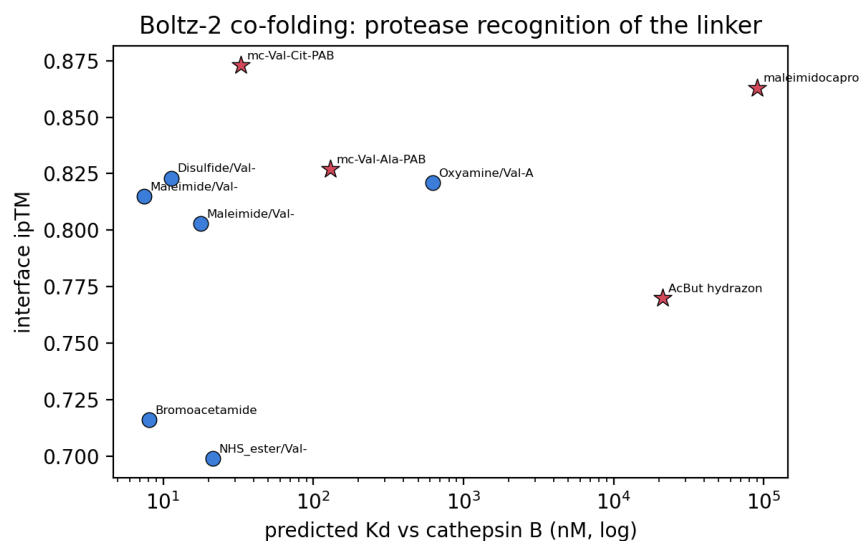


Figure 5: Boltz-2 cathepsin-B co-folding: interface confidence (ipTM) vs predicted Kd for designed linkers (blue) and commercial controls (red stars). Designed linkers cluster with the clinical substrate.